# Sahaj Bhadja

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## Summary

Software engineer with 2+ years building reliable cloud services, REST APIs, automated workflows, and production tooling. Experienced in Python, Java, React, AWS, Azure, observability, CI/CD, incident troubleshooting, testing, failure recovery, and operational automation.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Built Python APIs, Azure SQL services, and operational dashboards for 500+ users, profiling indexed queries and backend access paths to reduce production latency from 1.2 seconds to 350 milliseconds.
- Orchestrated 5,000+ monthly jobs through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving observability, operational resilience, recovery, maintainability, and systematic production incident troubleshooting responsiveness.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, integrating authentication, document workflows, status telemetry, and operational controls while debugging production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*
- Built testable Python and PyTorch pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, structured evaluation, performance monitoring, validation checks, and systematic model debugging.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

**Aasma Technology Solutions** **India**
*Software Developer Engineer – SecureSync* *August 2023 – July 2024*
- Developed Python services and REST integrations using filtering, pagination, caching, AWS Glue, S3, and SQL, improving response times 30% and data-processing efficiency 68% across production cloud applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating defect diagnosis, release verification, and reliable cross-platform production deployments across Windows, Linux, and macOS.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern – VibrantChem* *May 2023 – August 2023*
- Developed modular React and Node.js services across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, coordinating 2 million+ Azure SQL records through documented interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

## Skills

**Languages & Services:** Python, Java, JavaScript, TypeScript, SQL, React, Node.js, Flask, FastAPI, REST APIs, microservices, WebSockets, backend services
**Cloud & Reliability:** AWS, Azure, S3, AWS RDS, Azure SQL, Docker, Kubernetes, distributed workflows, retries, failure recovery, operational resilience, production support
**Observability & Operations:** logging, monitoring, dashboards, status telemetry, performance profiling, incident troubleshooting, failure diagnosis, operational debugging, state tracking, deployment automation
**Quality & Delivery:** Git, Linux, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, functional testing, automated testing, debugging, code review, documentation, build automation, Agile, Scrum

## Projects

**Big-Brain** | Live | *LLaMA-2, RAG, FAISS, PostgreSQL, REST APIs, Clerk*
- Built an authenticated RAG application processing 50,000+ unstructured documents with LLaMA-2, FAISS, PostgreSQL, and REST APIs, combining semantic retrieval, structured validation, access controls, and real-time collaboration workflows.

**Chat2SQL** | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular Java and TypeScript services.

**FinNexus** | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a responsive platform supporting 50,000+ users and one million transactions, integrating secure APIs, real-time WebSocket synchronization, MongoDB persistence, reusable interfaces, Kubernetes deployment, and automated release workflows.